

Sparsity-inducing regulariser for DNN involves the following:

L1 regulariser (sum of absolute values)

- It has been used for sparsity since 1996.
- It is differentiable, convex, and easy to optimise
- It is proportional to the scaling, which will lead to scaling down of all the elements with the same speed, which is undesired

L0 regulariser (number of nonzero values)

- It directly reflects the sparsity by definition
- It is scale invariant. However, our zero regulariser provides no useful gradients
- It needs additional tricks for applying on DNN pruning (stochastic approximation, ADMM), which makes the problem complicated. This leads to certain complexity in the design

Moving beyond lasso

We propose to move beyond L1 and L0 and aim to find the sparsity-inducing regulariser that is both differentiable and scale-invariant for pruning. At this stage the hoyer-square regulariser draws our attention. It is represented by the ratio between the L1 and L2, which is both differentiable and scale-invariant.

For element-wise DNN pruning, we can apply the square version of hoyer regulariser, which has the same range and similar minima structure as L0. We further extend the hoyer-square regulariser to the structure pruning settings.

We can apply hoyer-square regulariser over the L2 norm of each group of weight. The weight within the group can be induced to become zero simultaneously. Let us now look at the effect of the hoyer-square regulariser.

Regulating the activation sparsity:

In operations, the activation sparsity is dynamic and the sparsity level is not identical for different inputs. Our group has proposed what is called the Dynamic Winner Takes All (WTA) dropout

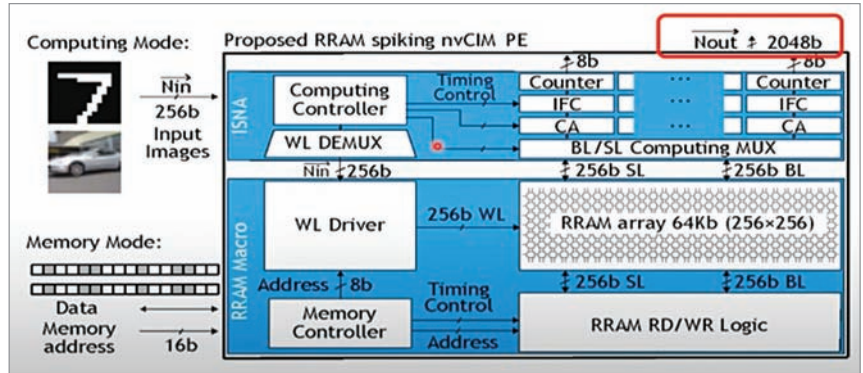


Figure 8: Processing engine with in-situ neuron activation

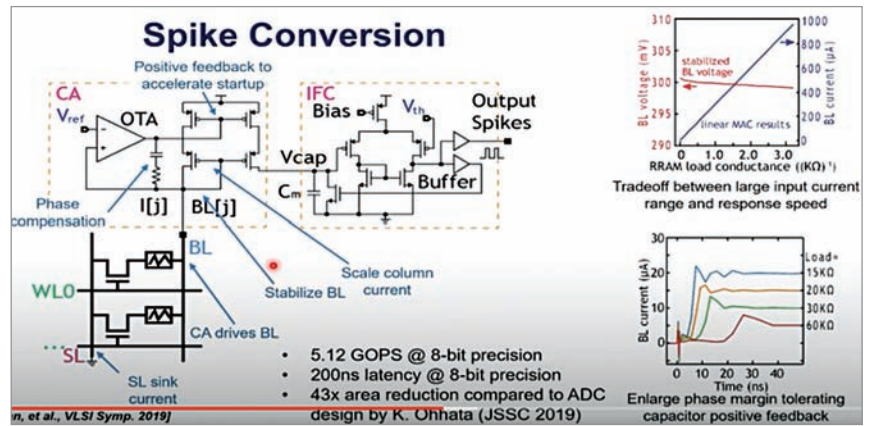


Figure 9: Spike conversion

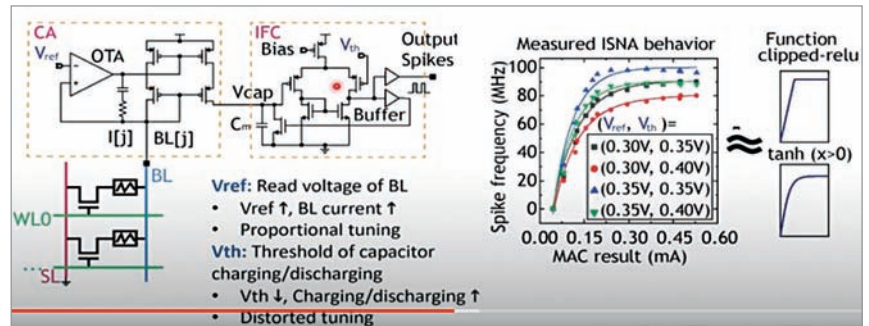


Figure 10: In-situ nonlinear activation function

technique to prune neuron activations layer wise. Hence, the design introduces low overhead for the real-time interface (see Figure 5).

Mixing precision with bit-level

sparsity: We know that fixed point quantisation is an important model compression technique. In the previous research it was found that some layers can retain higher precision while others can be quantised to less precision. This is the whole idea of mixed precision

quantisation.

For a quantised matrix, when the most significant bits of all the elements are zero simultaneously, we can remove the MSB directly (see Figure 6).

For bit representation we follow the dynamic scaling quantisation procedure. Here (see Figure 7) we use the element with the largest absolute value in each layer that is the scaling factor and uniformly quantise all the elements to embed the value in binary form.